

Test 9 — Answer Key

Each row gives the item number, the B.E.S.T. benchmark and difficulty, the correct answer, and a student-facing explanation.

#	Benchmark	Answer	Explanation
1	MA.6.AR.2.4 moderate	17	Multiply both sides by the divisor to find the unknown.
2	MA.6.NSO.3.2 challenging	$12(1 + 3)$	The GCF of 12 and 36 is 12, and $12 = 12 \times 1$, $36 = 12 \times 3$.
3	MA.6.NSO.1.2 moderate	6.0, -5.0, 0.0	A gain is a positive integer, a loss is a negative integer, and 0 means no gain or loss.
4	MA.6.GR.2.3 moderate	35	Multiply the three edge lengths together.
5	MA.6.NSO.3.3 challenging	16	2^4 means 2 multiplied by itself 4 times, not 2×4 .
6	MA.6.NSO.3.5 challenging	25%	Write the fraction as a decimal, then multiply by 100.
7	MA.6.GR.1.3 moderate	$(-6, 0)$	The fourth vertex lines up with A vertically and with C horizontally.
8	MA.6.NSO.3.2 moderate	12, 7, 5	Divide each addend by the GCF: $84 \div 12 = 7$ and $60 \div 12 = 5$.
9	MA.6.AR.1.4 moderate	40, 72	Distribute the factor to both terms inside the parentheses.
10	MA.6.NSO.3.1 moderate	GCF of 24 and 36 $\rightarrow$ 12; LCM of 24 and 36 $\rightarrow$ 72; GCF of 24 and 72 $\rightarrow$ 24	The GCF is the largest shared factor; the LCM is the smallest shared multiple.
11	MA.6.DP.1.4 challenging	6	Count one dot for each student at 6 and to the right of 6.
12	MA.6.GR.2.1 moderate	27	Multiply the base by the height, then take half.
13	MA.6.AR.3.4 challenging	80	40% means 40 out of every 100, so multiply the total by 40/100.
14	MA.6.NSO.2.3 moderate	\$24.16, \$6.04	Subtract the cost of the pass, then divide what is left by the number of people.
15	MA.6.AR.2.2 moderate	52	Add the same number to both sides to undo the subtraction.
16	MA.6.NSO.2.1 moderate	8.625, 138	Multiplying by a number less than 1 makes the result smaller; dividing by it makes the result larger.

#	Benchmark	Answer	Explanation
17	MA.6.DP.1.1 moderate	How many text messages did each student send yesterday?, How many minutes did each sixth grader spend on the bus this morning?	Statistical questions anticipate variability in the answers.
18	MA.6.NSO.1.1 moderate	-2.5, -1.75, -1.0	Each small tick mark is one fourth of a unit, so count fourths from 0 to place each number.
19	MA.6.NSO.4.2 challenging	-108	The product of two numbers with the same sign is positive; with different signs it is negative.
20	MA.6.NSO.1.4 moderate	-35.0, 35.0	Depths of -35 and 35 feet are both 35 units from 0.
21	MA.6.DP.1.5 challenging	2, 6 $\frac{1}{2}$, 23, 27 $\frac{1}{2}$, 29	Order the data, find the median, then find the medians of the lower and upper halves.
22	MA.6.DP.1.6 challenging	19 $\frac{1}{2}$	Add the new value to the total and divide by the new number of values.
23	MA.6.NSO.1.3 moderate	3.0, -3.0	Two numbers are 3 units from 0: 3 and -3.
24	MA.6.NSO.3.4 challenging	$2^2 \times 3^2 \times 11$	Break the number into factors until every factor is prime, then use exponents.
25	MA.6.AR.1.1 moderate	$x + 6$	Match each word in the description to an operation in the expression.
26	MA.6.AR.2.3 moderate	44	Multiply both sides by the divisor to undo the division.
27	MA.6.DP.1.2 moderate	18	Add the five values and divide the total by 5.
28	MA.6.NSO.2.2 moderate	$\frac{5}{6} \times \frac{4}{9} \rightarrow \frac{10}{27}$; $\frac{5}{6} \div \frac{4}{9} \rightarrow \frac{15}{8}$; $\frac{4}{9} \div \frac{5}{6} \rightarrow \frac{8}{15}$	Multiply straight across; to divide, multiply by the reciprocal.
29	MA.6.GR.2.2 moderate	100	Find the area of the whole rectangle, then subtract the corner that is cut out.
30	MA.6.GR.2.4 moderate	114	Find the area of each face from the net and add all six.
31	MA.6.AR.2.1 moderate	12	An equation is true only for the value that balances both sides.
32	MA.6.AR.3.2 challenging	\$6.00	Divide the total cost by the number of pounds to get the cost for one pound.
33	MA.6.AR.3.3 challenging	14	Each row multiplies the first ratio by the same number, here 2.
34	MA.6.NSO.4.1 challenging	28	Adding a negative number moves left on the number line; adding a positive number moves right.

#	Benchmark	Answer	Explanation
35	MA.6.AR.1.2 moderate	61, 65, 69	Test each value in the inequality and keep the ones that make it true.
36	MA.6.AR.1.3 moderate	15	Multiply the rate by the miles, then add the base fare.
37	MA.6.AR.3.5 challenging	180 inches	Each foot is 12 inches, so multiply the number of feet by 12.
38	MA.6.GR.1.1 moderate	$(-2, 5)$, $(2, 5)$, $(-2, -5)$	Start at the origin, move along the x-axis first, then up or down.